Improving Diffusion Policies with Latent Noise Steering for Planar Manipulation

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Abstract

Behavior cloning of human demonstrations provides an effective way to acquire contact-rich manipulation skills without relying on heavy model predictive control, which can be sensitive to model inaccuracies and computationally expensive for highly nonlinear, discontinuous dynamics. In this work, we study behavior cloning for non-prehensile planar manipulation, where a robot must move objects through contact without grasping. We focus on the Push-T task, a benchmark requiring a robot arm to push a T-shaped object to a desired pose on a tabletop. The multimodal nature of this task—where humans naturally demonstrate diverse solution modes—motivates the use of denoising diffusion probabilistic models (DDPMs) as the policy class. Diffusion policies can represent complex, multimodal action distributions and generate smooth contact-rich behaviors. We show that a diffusion-based behavior cloning policy can successfully learn human manipulation strategies and generalize to different start and goal poses while mantaining kinematic feasibility. To improve the policy beyond behavior cloning, we apply a paradigm called Diffusion Steering via Reinforcement Learning, which learns an auxiliary noise policy from which the initial noise in the DDPM is sampled, trained via actor-critic methods. We observe moderate improvement in our multi-goal policy, showing that RL finetuning can help behavior cloned diffusion policies generalize with limited demonstration data.

1 Introduction

Robotic manipulation in unstructured environments often requires interacting with objects through complex contact dynamics. While grasping provides a reliable means of securing objects, many real-world tasks demand *non-prehensile manipulation*—pushing, sliding, or rolling objects without establishing a stable grasp. Such behaviors are essential when objects are too large, heavy, or geometrically awkward to grasp, or when the robot is equipped with simple end-effectors. However, non-prehensile interactions are challenging to model: the resulting dynamics are highly nonlinear, discontinuous, and sensitive to small variations in contact conditions such as friction or surface geometry.

A common strategy for tackling contact-rich manipulation is to use model predictive control (MPC) with analytical or learned dynamics models. Despite significant progress, MPC approaches face two persistent difficulties. First, their performance depends critically on model accuracy, which is difficult to guarantee for contact dynamics. Second, online trajectory optimization can be computationally demanding and prone to getting trapped in poor local minima, especially in tasks with many feasible solution modes. These limitations impair real-time deployment and robustness in settings where repeated contact events occur.

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An alternative is to bypass explicit modeling entirely and directly learn a policy from human demonstrations. *Behavior cloning* (BC) provides a straightforward and scalable pathway for distilling human expertise into robot behavior, particularly for tasks requiring nuanced, contact-rich strategies that humans perform naturally. However, standard BC methods often struggle with multimodality: when multiple successful strategies exist, unimodal policies such as MLPs or Gaussian regressors tend to average disparate demonstrations, producing physically invalid actions. This challenge is especially apparent in non-prehensile manipulation tasks where humans may push, pivot, scoop, or rotate objects in qualitatively distinct ways.

In this work, we examine these issues through the *Push-T* task, which requires a robot arm to move a T-shaped rigid object on a tabletop from an initial to a target pose. Push-T is inherently multimodal—solvers may choose to push on different edges, rotate the object at intermediate poses, or approach from multiple directions. Human demonstrations naturally reflect this diversity, making Push-T an ideal testbed for evaluating policy classes that must capture multimodal behavior.

To address these challenges, we propose using *denoising diffusion probabilistic models* (DDPMs) as a behavior cloning policy. Diffusion models learn a generative distribution over demonstrated trajectories and can represent multiple solution modes without collapsing to a mean strategy. At inference time, the model samples from a conditional posterior over trajectories given recent observations, allowing it to generate coherent, contact-rich motion sequences without explicit dynamics modeling.

Our contributions are as follows:

- We demonstrate that diffusion-based behavior cloning provides a robust and expressive policy class for non-prehensile manipulation, capable of capturing multimodal strategies from human demonstrations.
- We develop and evaluate a diffusion policy for the Push-T benchmark, showing that it reproduces human-like solution modes that standard unimodal policies fail to capture.
- We analyze the benefits of generative trajectory modeling in contact-rich tasks and show that diffusion policies can generalize to different start and goal poses while maintaining kinematic feasibility.
- We train an auxilary noise policy with Diffusion Steering via Reinforcement Learning (DSRL) and evaluate performance improvement on our multi-goal behavior cloning diffusion policy.

By leveraging the multimodal expressiveness of diffusion models, our approach offers a scalable and model-free framework for learning complex non-prehensile manipulation behaviors, without relying on computationally expensive trajectory optimization or brittle analytical contact models. Additionally, finetuning behavior cloned policies with DSRL is an innovative technique to improve performance while only assuming black box access to the original policy, which can enable efficient online adaptation without collecting more human demonstrations.

2 Related Work

2.1 Non-Prehensile Manipulation

Non-prehensile manipulation encompasses a broad class of strategies in which robots interact with objects through intermittent or sustained contact without grasping, including pushing, sliding, pivoting, and rolling. Prior work has investigated analytical and data-driven approaches for modeling contact dynamics, planning, and control in such settings, often highlighting the inherent challenges arising from discontinuous frictional interactions and underactuated object motion. Classical methods rely on quasi-static approximations, motion cones, or hybrid dynamics models to reason about feasible pushes, whereas more recent learning-based approaches leverage demonstrations or reinforcement learning to bypass explicit modeling and acquire robust manipulation skills directly from experience. Despite progress, capturing the multimodal nature of human manipulation strategies remains difficult, motivating expressive policy classes capable of representing diverse behaviors.

2.2 Diffusion Models

Denoising diffusion probabilistic models (DDPMs) have emerged as powerful generative models capable of representing complex, multimodal data distributions. Diffusion models construct a forward noising process that gradually corrupts data with Gaussian noise, and learn a reverse process that incrementally denoises samples to recover clean data. This iterative denoising procedure enables flexible sampling and stable training, and has been shown to capture richer distributions than variational autoencoders or GAN-based approaches. In robotics and reinforcement learning, diffusion models have been explored as expressive trajectory and action generators, enabling planning or policy learning in domains with highly multimodal solution spaces [1].

2.3 Diffusion Policies for Manipulation and Reinforcement Learning

Recent work has applied diffusion models specifically to robotic manipulation, leading to the development of *diffusion policies* that learn distributions over state-conditioned actions or trajectories. Diffusion policies have been shown to outperform unimodal policy classes in offline reinforcement learning and imitation learning settings, particularly in tasks involving contact-rich interactions or multiple valid solution modes [2]. Extensions include value-guided denoising, which integrates critic information into the sampling process to bias the diffusion model toward higher-value behaviors [3]. Comprehensive surveys highlight the rapidly growing ecosystem of diffusion-based manipulation methods and emphasize their advantages in behavior expressiveness, multimodal action prediction, and robustness to distributional complexity in human demonstrations [4]. These capabilities make diffusion models a natural fit for non-prehensile manipulation tasks where humans demonstrate diverse strategies.

3 Methods

3.1 Push-T Environment and Task Setup

We study a non-prehensile manipulation task in which a Franka Emika Panda arm, equipped with a rigid rod attachment at the end-effector, must push and reorient a T-shaped rigid object on a tabletop. The objective is to move the object from an initial SE(2) pose to a desired goal pose. Let the T-object's pose be represented as

$$T = (x_T, y_T, \theta_T), \qquad T_g = (x_g, y_g, \theta_g), \tag{1}$$

where (x, y) denotes planar position and θ the yaw orientation.

A demonstration is considered successful when a set of predefined keypoints on the T-object lie within an ℓ_2 tolerance of the goal pose:

$$\|p_i - p_i^{(g)}\|_2 \leq \epsilon, \quad \forall i, \tag{2}$$

where p_i and $p_i^{(g)}$ denote corresponding keypoints on the current and goal configurations.

This task admits multiple feasible strategies, such as pivot pushes, side pushes, and edge rotations, making it a suitable benchmark for multimodal policy learning.

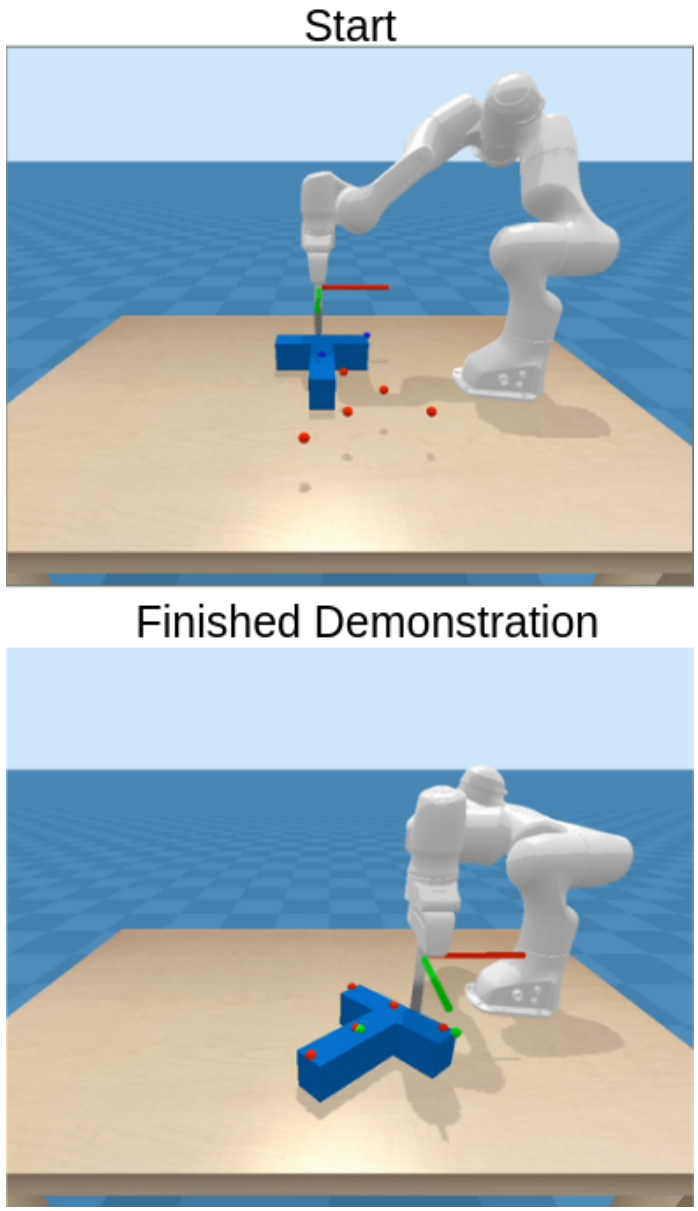

Figure 1: Demonstration is considered successful once key points on the T lie within a predefined L2 distance of the goal position.

3.2 Observation and Action Spaces

At each timestep, the observation consists of:

- the end-effector Cartesian position $e = (x_e, y_e, z_e)$,
- the full SE(3) pose of the T-object (x_T, y_T, z_T, q_T),
- the full SE(3) goal pose (x_g, y_g, z_g, q_g),

where q_T and q_g denote quaternion orientations.

Actions are specified in end-effector space as desired position displacements

$$a_t = \Delta(x, y, z), \tag{3}$$

which are converted to joint-space targets via inverse kinematics:

$$q_t^* = \mathrm{IK}(e_t + a_t). \tag{4}$$

A joint-space PD controller then executes

$$\tau_t = K_p(q_t^* - q_t) + K_d(\dot{q}_t^* - \dot{q}_t), \tag{5}$$

ensuring stable tracking of the desired end-effector motion.

Consistent with prior observations in diffusion policy literature [5], we found that Cartesian position actions yield more stable training and higher-quality rollouts than velocity-based actions.

3.3 Human Demonstration Collection

Expert demonstrations are collected using a joystick-based teleoperation system that maps joystick inputs to Cartesian end-effector commands. The operator manipulates the rod-equipped end-effector to push the T-object from its initial pose to the goal pose. Each trajectory contains the full sequence of observations and actions over a fixed horizon.

Two datasets are collected:

Dataset 1 (Structured Instances) The start and goal poses of both the end-effector and T-object are fixed across demonstrations. This reduces variability and produces consistent strategies.

Dataset 2 (Randomized Instances) The initial and goal poses are randomized within a bounded workspace:

$$(x_T, y_T), (x_g, y_g) \in \mathcal{W}, \qquad \theta_T, \theta_g \in [-\pi, \pi], \tag{6}$$

resulting in diverse pushing strategies and significantly more multimodal behaviors.

Each dataset contains approximately 100 successful demonstrations.

3.4 Diffusion Policy Formulation

3.4.1 Denoising Diffusion Probabilistic Model

Denoising diffusion probabilistic models (DDPMs) [6, 7] define a generative model by inverting a fixed forward noising process. Let τ_0 denote a clean data sample (e.g., a state–action trajectory segment). The forward diffusion process gradually corrupts the data via:

$$q(\tau_i \mid \tau_{i-1}) = \mathcal{N}(\tau_i \mid \sqrt{\alpha_i}\,\tau_{i-1}, (1-\alpha_i)I)\,, \tag{7}$$

where $\alpha_i \in (0, 1)$ controls the variance schedule. After N steps, τ_N approaches an isotropic Gaussian distribution.

The model learns a reverse denoising process

$$p_\theta(\tau_{i-1} \mid \tau_i), \tag{8}$$

which reconstructs τ_{i-1} from a noisier sample τ_i. The joint generative model is given by

$$p_\theta(\tau_0) = \int p(\tau_N) \prod_{i=1}^{N} p_\theta(\tau_{i-1} \mid \tau_i)\, d\tau_{1:N}, \tag{9}$$

where $p(\tau_N) = \mathcal{N}(0, I)$ is the Gaussian prior. In practice, the reverse conditionals are parameterized as Gaussians with fixed, time-dependent covariance:

$$p_\theta(\tau_{i-1} \mid \tau_i) = \mathcal{N}(\tau_{i-1} \mid \mu_\theta(\tau_i, i), \Sigma_i)\,. \tag{10}$$

The parameters θ are learned by minimizing a variational lower bound on the negative log-likelihood:

$$\theta^\star = \arg\min_\theta -\mathbb{E}_{\tau_0}\big[\log p_\theta(\tau_0)\big]. \tag{11}$$

3.4.2 Noise Prediction Loss

Training proceeds by sampling a clean data point x_0 from the demonstration dataset, selecting a random diffusion step k, and corrupting x_0 using the analytic forward noising process:

$$x_k = \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1 - \bar{\alpha}_k}\, \varepsilon_k, \tag{12}$$

where $\varepsilon_k \sim \mathcal{N}(0, I)$ and $\bar{\alpha}_k = \prod_{i=1}^{k} \alpha_i$.

The noise prediction network ε_θ is trained to predict the injected noise:

$$\mathcal{L}_{\text{MSE}} = \text{MSE}(\varepsilon_k,\ \varepsilon_\theta(x_k, k))\,. \tag{13}$$

As shown by Ho et al. [7], this simple denoising objective is equivalent to optimizing the variational lower bound on $\text{KL}(p(x_0) \,\|\, q_\theta(x_0))$ for the generative model. Thus minimizing the MSE noise-prediction loss directly trains the DDPM to reconstruct clean trajectories from noise-corrupted versions.

3.4.3 Transformer-Based Diffusion Policy Architecture

To capture long-range temporal dependencies in manipulation trajectories, we adopt a transformer-based diffusion architecture. At diffusion iteration k, the noisy action sequence A_t^k is provided as a sequence of input tokens to a stack of transformer decoder blocks. A sinusoidal embedding of the diffusion timestep k is prepended as the first token, enabling the model to condition its denoising behavior on the noise level.

The observation O_t is processed through a shared MLP encoder to produce an observation embedding sequence, which is fed into the transformer decoder as context features. The denoising network then predicts the noise associated with each action token:

$$\varepsilon_\theta(O_t, A_t^k, k), \tag{14}$$

where each output token of the decoder corresponds to the noise estimate for a single action dimension at time t. This architecture allows the diffusion policy to jointly model temporal structure, multimodal action distributions, and observation-conditioned behavior, making it well suited for contact-rich manipulation tasks.

3.5 Diffusion Steering via Reinforcement Learning

Vanilla diffusion policies generate actions at test time by iteratively denoising from Gaussian noise using the learned reverse process. By changing the parameters of the Gaussian which we sample noise from, the policy will produce different trajectories. When demonstrations are suboptimal, sparse, or misaligned with the evaluation objective, vanilla diffusion policies can be brittle. In addition, when we encounter goals at inference time that were unseen in the demonstrations, we have no control over the resulting performance. The main objective of Diffusion Steering via Reinforcement Learning (DSRL) [8] is to optimize the denoising process by learning a policy π_{dp}^w that actively steers the generative process toward high-reward trajectories by altering the input noise distribution. The action space of this policy is the latent-noise space, $w \in \mathcal{W}$, and diffusion policy actions are produced by denoising using the DSRL policy actions, $a \leftarrow \pi_{dp}^w(s, w)$. This now forms a latent-action MDP, in which we substitute the diffusion policy level actions with the denoising result of the DSRL policy:

$$P^{\mathcal{W}}(\cdot|s, w) := P(\cdot|s, \pi_{dp}^w(s, w)), \quad \mathcal{M}^{\mathcal{W}} := (\mathcal{S}, \mathcal{W}, P^{\mathcal{W}}, p_0, r^w, \gamma) \tag{15}$$

DSRL allows us to effectively treat the diffusion policy as a black box since training a DSRL policy does not require backpropagating through the gradients of the base policy, and only requires forward passes of the diffusion policy in order to optimize π_{dp}^w. To improve the sample efficiency and learnability of DSRL, we introduce a noise-aliased variant of the algorithm that learns two critic functions. The first critic, $Q^{\mathcal{A}}$, maps denoised diffusion actions to expected discounted returns. The second critic, $Q^{\mathcal{W}}$, maps latent noise inputs to expected discounted returns.

In diffusion policies, many distinct latent noise vectors can produce the same denoised action, which induces a highly aliased and non-smooth value landscape when learning directly in noise space. To mitigate this issue, we train $Q^{\mathcal{A}}$ using standard temporal-difference targets defined over denoised actions, and train $Q^{\mathcal{W}}$ by regressing its outputs to those of $Q^{\mathcal{A}}$ for the corresponding denoised actions (Eq. 16-17). This decomposition transfers a smoother learning signal from action space to noise space, substantially improving critic stability and learnability. We also conduct an ablation

called DSRL-SAC, which only utilizes a single critic that maps latent noise to Q-values.

$$Q^{\mathcal{A}}: \quad \min_{Q^{\mathcal{A}}} \mathbb{E}_{(s,a,r,s')\sim\mathcal{B},\, a'\sim\pi^{\mathcal{W}}_{\mathrm{dp}}(s',\pi^{\mathcal{W}}(s'))}\Big[\big(Q^{\mathcal{A}}(s,a) - r - \gamma\,\bar{Q}^{\mathcal{A}}(s',a')\big)^2\Big], \tag{16}$$

$$Q^{\mathcal{W}}: \quad \min_{Q^{\mathcal{W}}} \mathbb{E}_{s\sim\mathcal{B},\, w\sim\mathcal{N}(0,I)}\Big[\big(Q^{\mathcal{W}}(s,w) - Q^{\mathcal{A}}\big(s,\pi^{\mathcal{W}}_{\mathrm{dp}}(s,w)\big)\big)^2\Big]. \tag{17}$$

4 Experiments and Results

We first validate our diffusion policy architecture on standard offline reinforcement learning benchmarks from the D4RL suite, focusing on the PointMaze environments, which require long-horizon reasoning and precise navigation through narrow passages. In this domain, the agent controls a force-actuated ball whose motion must be coordinated over extended trajectories to avoid dead-ends and reach the goal efficiently. The expert data is from a Q-learning agent equipped with a PD controller for stabilizing low-level dynamics, resulting in consistent and high-quality demonstrations. Our diffusion policy successfully models the multimodal structure of the expert trajectories and reproduces long-horizon behaviors characteristic of the task, which serves as a reasonable baseline before approaching the more complicated manipulation task.

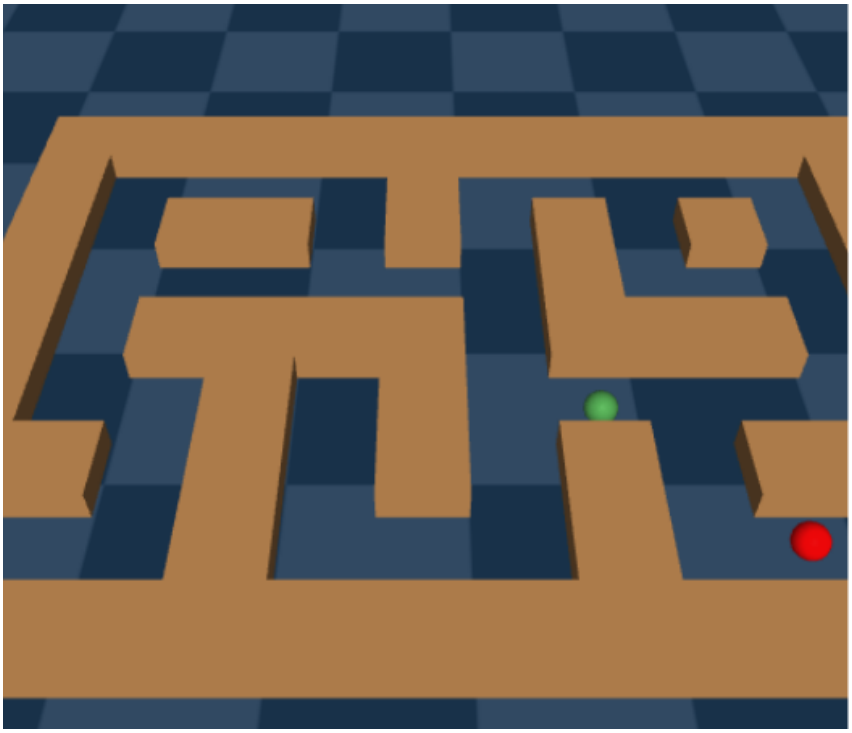

Figure 2: PointMaze_Large-v3 Environment from D4RL

To train our transformer-based diffusion policy on our teleoperated datasets of 100 state action trajectories, we use 50k training iterations for single goal and 70k for multi-goal, with a batch size of 256, conditioned on 9 previous actions and 10 previous states for denoising. At the end of training, our MSE loss (13) plateaus around 0.003 for both policies.

Table 1: Single-goal vs. multi-goal experiment results over 100 episodes.

Experiment	Avg. Reward	Median Reward	Max Reward
Single Goal Policy	-71.07	-70.94	-58.92
Multi Goal Goal	-131.51	-134.68	-29.59

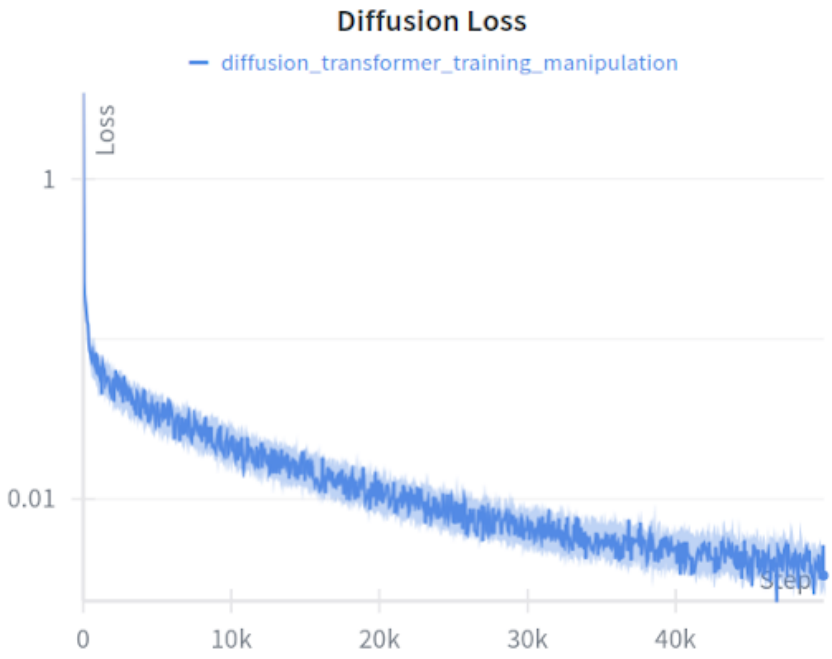

Figure 3: Training loss of single goal diffusion policy

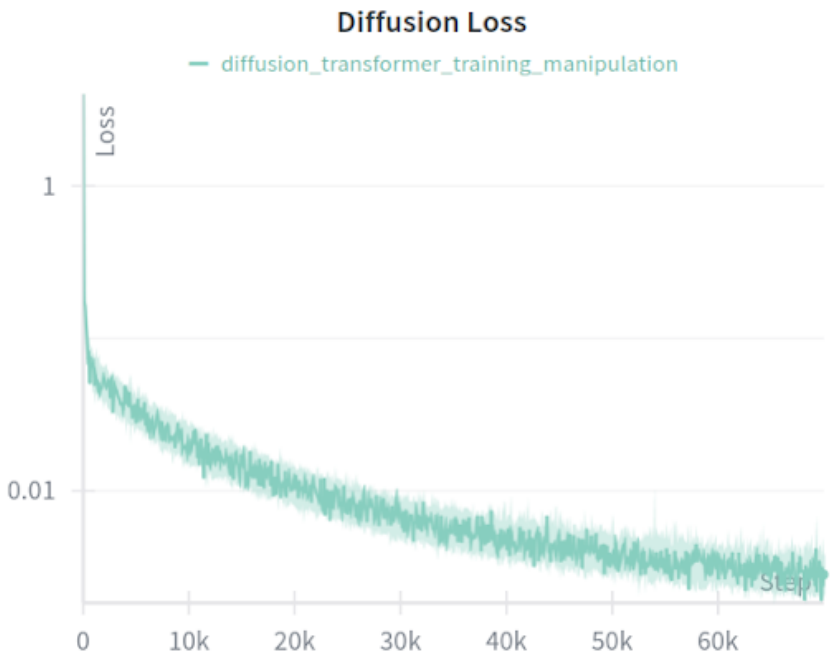

Figure 4: Training loss of multi goal diffusion policy

Table 2: Performance Comparison of Vanilla Diffusion to DSRL Methods, Multi-Goal, 100 Evaluation Episodes

	Vanilla Diffusion	**DSRL-SAC**	**DSRL-NA**
Avg. Reward	-131.51	-141.78	-105.95

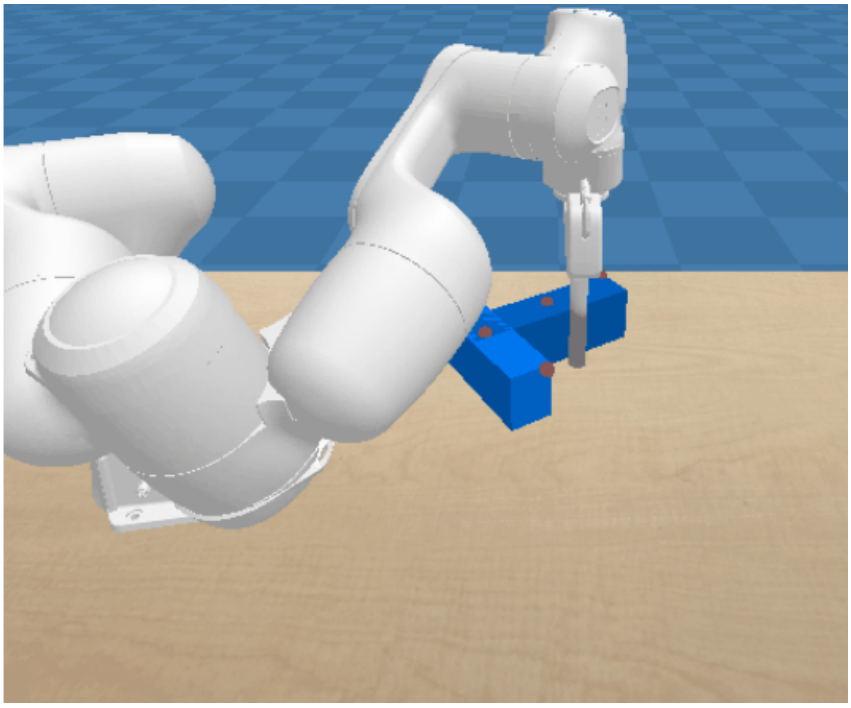

Figure 5: Policy successfully completes reorientation task

To train our DSRL policy, we use 100k training steps with a batch size of 256. Our actor and critic networks are three MLP layers of size 2048, and we roll out for 2000 steps of standard diffusion to collect data for the replay buffer prior to training. We trained both DSRL-NA and DSRL-SAC, and found that the noise aliased version of the algorithm significantly outperformed the single-critic version. DSRL-NA had issues with loss estimates, with both actor and critic losses steadily increasing over the course of training instead of converging to a minimum value. DSRL-SAC, conversely, showed converging loss over the course of training. Despite the qualitatively better loss plots, DSRL-SAC was still unable to perform as well as DSRL-NA, suggesting that DSRL-NA tends to converge to very suboptimal minima and enforcing the belief that the direct noise to Q-value mapping is very difficult to learn.

To evaluate the learned diffusion policy, we deploy it on a set of Push-T manipulation tasks spanning a variety of start and goal object poses. When the initial configuration of the T-object and the target pose lie within the distribution of demonstrations seen during training, the policy reliably generates coherent pushing behaviors and successfully reorients the object to the goal pose. However, when either the start or goal pose deviates substantially from the training distribution—such as requiring large rotations or translations not present in the demonstrations—the policy struggles to produce effective contact sequences and often fails to reach the desired configuration. This limitation reflects the dataset-dependent nature of behavior cloning and the difficulty of extrapolating multi-modal contact strategies beyond demonstrated regimes. As expected, we observe higher success on the single-start, single-goal setting, where test-time configurations exactly match those encountered during training, compared to the multi-start, multi-goal setting, which requires broader generalization across diverse object placements.

5 Conclusion

In this work, we presented a diffusion-based behavior cloning framework for learning non-prehensile planar manipulation strategies from human demonstrations. By leveraging the generative expressiveness of denoising diffusion probabilistic models and a transformer-based denoising architecture, our method effectively captures the multimodal nature of human pushing behaviors in the Push-T task. Through experiments on both standard offline RL benchmarks and teleoperated manipulation datasets, we demonstrated that diffusion policies can reproduce smooth, contact-rich trajectories and achieve reliable performance when evaluated on start and goal configurations that lie within the demonstrated data distribution. At the same time, our results highlight the inherent challenge of extrapolating beyond demonstrated regimes: performance degrades when the initial or target object pose deviates significantly from the dataset, reflecting a core limitation of purely demonstration-driven methods. While our DSRL finetuning framework shows some improvement with these challenges, more experiments need to be done to find optimal strategies and architecture choices for reinforcement learning.

Looking forward, several promising directions may further extend the capabilities of diffusion-based manipulation policies. First, integrating our approach with goal-conditioned reinforcement learning techniques—such as hindsight experience replay or advantage-weighted methods—may help overcome the long-horizon reasoning challenges observed in out-of-distribution settings. Using an approach like DSRL allows for fine-tuning of large pretrained diffusion models purely with black box access, negating the need to retrain base diffusion models for new task objectives. This is analogous to how LLMs can be fine-tuned to specialize without retraining the base model, and could provide a path to generalist diffusion model backbones trained on many different tasks and environments. Exploring these directions offers a path toward more robust, scalable, and adaptive manipulation policies that combine the strengths of generative modeling with reinforcement learning.

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